Materials Resource Management

Universität Augsburg University

THESIS TYPE

The title

The author

for obtaining the academic degree *Degree*
in the course *Program*.

Materials Resource Management

Universität Augsburg University

University of Augsburg
Faculty of Mathematics, Natural Sciences, and Materials Engineering
Professorship

Supervisor:	The supervisor
First examiner:	The first examiner
Second examiner:	The second examiner

Augsburg, May 2023

Declaration

I affirm that the thesis was compiled by myself and using exclusively the sources and materials cited.

The author, The City, May 2023

Acknowledgments

This is an opportunity to thank people supporting your thesis, e.g. technicians, supervisors, friends providing mental support, etc.

Abstract

An abstract is a condensation of the content into one paragraph, at most one page. The abstract should stand on its own and present the motivation, methods, goals, and results of the work to someone who does not have time to read the entire work. It must enable a reader to decide whether he should read this work. The abstract is of great importance and should be formulated very consciously, because crucial key points of the work must become clear.

Context: First, the reader should get on board. For this purpose, one or two sentences are used to introduce the subject.

State of research: At this point, the current state of research is summarized very briefly. In particular, shortcomings of previous results should be addressed in order to work towards one's own research question.

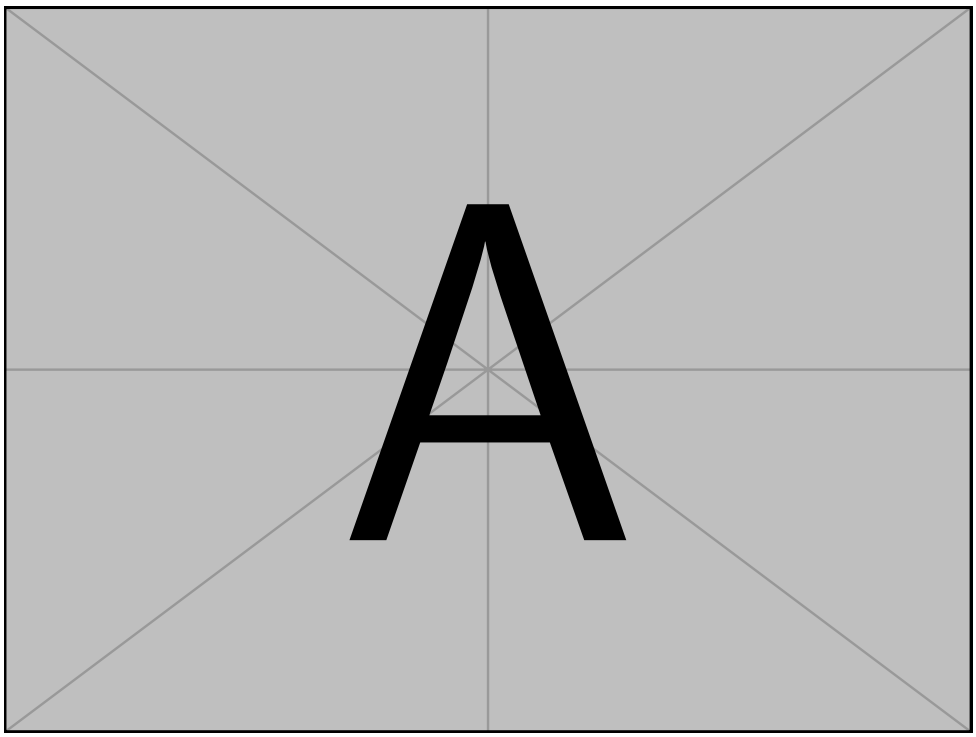
Research gap: Here the own research question is worked out. What is missing so far or which aspects need to be examined more closely?

Own approach: This section describes the methods used - what is one's approach to solving the research question?

Results: What came out of the work and which conclusions may be drawn?

Graphical abstract

You may summarize your work in a graphical abstract. Here is some more info on graphical abstracts: www.elsevier.com/authors/tools-and-resources/visual-abstract



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Acronyms

CMC	Ceramic Matrix Composites.
SVD	Singular Value Decomposition.

Symbols

Symbol	Description	Unit
$\mathbb{C}$	Stiffness tensor	GPa
π	Geometrical constant for circles	-

1 Introduction

This chapter should start with the motivation outlining the necessity and novelty of the scientific endeavor. The goal of the introduction is the elaboration of the research question from which the objective of the thesis is derived. The specific aspects which are examined are to be illustrated in a brief and concise manner. This chapter, as well as the abstract, can be finalized towards the end of the thesis in order to ensure the coherence with the last chapter (conclusion).

2 State of research / Literature review / Background / Research gap / Objective

This chapter is supposed to prepare the reader for the following chapters. It includes the analysis of relevant scientific work on the given topic (state of research) as well as the elaboration of the fundamental knowledge necessary for the presentation of one's own work. While writing, assume that the reader has a professional background without a deep knowledge within the specific field of research. The gap within current research activities should be identified leading to the justification of your thesis work.

3 Methodology / Modeling / Experimental Methods / Experimental Procedure

This chapter should describe the applied methods or the newly developed methods to obtain research results. There is a difference between an experimental thesis and a modeling thesis in this chapter. The first one will have a rather descriptive character, while the second one will include more own developments in this chapter.

4 Results and discussion

In this chapter you should compile the results of your work into graphs and tables showing relationships within investigated parameters. In the discussion you should critically interpret and cross-link your results taking into account the current state. You should also comment on the reliability of your results.

5 Conclusion and outlook

This is a short chapter summarizing the key findings of your work. In addition, it should provide an outlook on further steps in the research field.

This chapter should show a link to the issues and gaps highlighted in the introduction. Introduction and conclusion form a key part of a good thesis to motivate reading your work.

Example chapter

Citations

This chapter will make use of many citations, hence some notes on citing are summarized here. There are two options to cite in LaTeX:

1. The command `\cite{key}` creates an automatically updated reference number, such as in this example: *This is a sample citation [1]*.
2. The command `\citet{key}` adds the author name, such as in this example: *Einstein [1] wrote an important paper*. One author or two authors should be explicitly named, more authors are denoted *Einstein et al.* . The command `\citet{key}` takes care of this automatically.

The keys and bibliographic data for the citations are listed in the file `literature.bib`. This can be created manually using the *BibTex* format or by any literature management software. Usage of such a literature management software (e.g. Citavi, Mendely, ResearchRabbit, or Endnote) is highly recommended. The bibliography should contain just references, which are actually used and referred to. In the default configuration of this template, the template takes care of this automatically.

Images, tables, technical specifications, etc. have to be cited as well. Adaptions of a figure from literature should be indicated as well, e.g. by writing *adapted from ....*

Language and style

A scientific thesis requires concise and precise language. Using past tense is reasonable in this section, as other people may have contributed to the state of research in the past, but apart from this, present tense should be used throughout the work. Expressions like *high temperature*, *good rheology* or every-day language should be avoided.

Use sections and subsections reasonable and consistent. It should not be too granular and subsections do not make any sense, if there is not at least one other subsection at

the same level. It is an important task of a student thesis to find a good structure and should be discussed in close coordination with supervisors.

Figures

This chapter will probably make use of several figures, e.g. showing experimental setups or sketches. The subsequent code showcases insertion of an image, where the expression *path* should be replaced with the actual path to your image.

```
\begin{figure}[ht]
    \centering
    \includegraphics[width=0.75\textwidth]{path}
    \caption{A caption describing the image.}
    \label{fig:image_label_a}
\end{figure}
```

This sample code results in the image shown in Figure 5.1. Please note that references to figures are capitalized, e.g. `Figure \ref{fig:image_label_a}`. Every image must be referenced in the text, there is no point having an image that is never mentioned in the text. Preferable, you should refer to the image before it appears in the document.

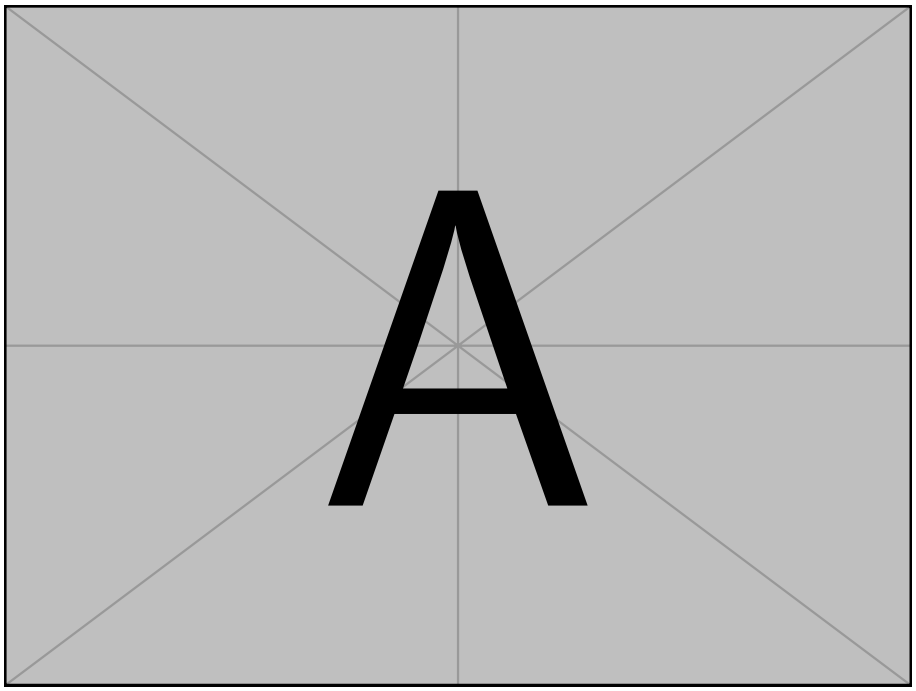


Figure 5.1: A caption describing the image. The caption should stand on its own and include a reference, if the image is adapted from another author.

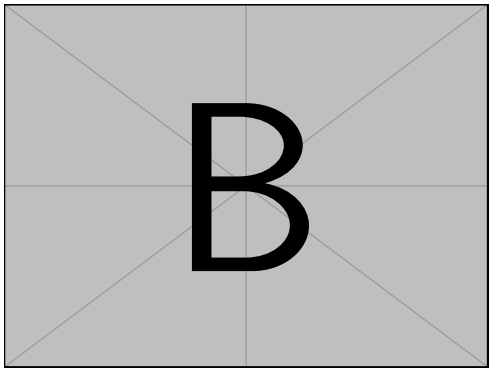
Make sure that the image is readable, i.e. features are large enough and the image quality is high. For pictures, use `JPEG` format, for diagrams and sketches use `PNG` format and for diagrams its best to use `PDF` format for optimal trade-offs between file size and quality. For diagrams, make sure to include axis labels with a sufficiently large font and use *in* <*unit*> instead of rectangular brackets to indicate units. Make sure that colors are easy to distinguish, some background on colors may be found https://colorbrewer2.org and https://bids.github.io/colormap/.

We may also combine multiple images, for example by the following code

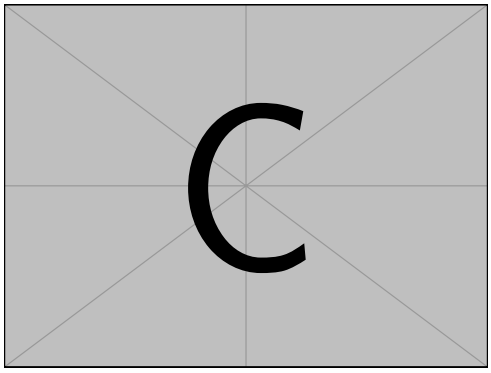
```
        \begin{figure}[ht]
         \centering
         \begin{subfigure}[b]{0.4\textwidth}
             \centering
             \includegraphics[width=\textwidth]{example-image-b}
             \caption{Sample image B}
             \label{fig:y equals x}
         \end{subfigure}
         \hfill
         \begin{subfigure}[b]{0.4\textwidth}
             \centering
             \includegraphics[width=\textwidth]{example-image-c}
             \caption{Sample image B}
             \label{fig:three sin x}
         \end{subfigure}
        \caption{Two images side by side}
        \label{fig:two_figs}
    \end{figure}
```

which results in Figure 5.2. This is useful for similar images or images comparing similar information.

In diagrams with nominal features, i.e. they do not have a natural order such as *bananas*, *apples* and *peaches*, bar charts should be used. Never interpolate such data with any lines. If data is obtained from experiments, information about scatter must be provided, e.g. with box plots showing percentiles or at least by providing a standard deviation for Gaussian processes.



(a) Sample image B



(b) Sample image B

Figure 5.2: Two images side by side

Tables

Tables are included inside via the environment `table`. There are several ways to create the actual table. The most common are the `tabular*` and `tabularx` environments. An example of a table can be found below in Table 5.1.

Table 5.1: Headline of the example table

	Thickness t in mm	**Young's modulus** E in GPa	**Strength** R_m in MPa	**Density** ϱ in $\mathrm{kg\,m^{-3}}$
Material 1	100.0	2000	1500	3.0
Material 2	2.0	3000	880	55.0
Material 3	0.6	4	713	52.8

Tables must have headlines and must be referred to in the text. Numbers should be aligned to the right and vertical lines should be omitted.

Equations

Important formulas should be inserted via the `equation` environment, which numbers them and offers the option to refer to equations via `\eqref`. For example, the code

```
\begin{equation}
    f(x)  = \sin \left(\frac{\sin(x)}{x} \right)
        \label{eq:key}
\end{equation}
```

results in

$$f(x) = \sin\left(\frac{\sin(x)}{x}\right). \tag{5.1}$$

Note that equations are part of the text and proper punctuation is needed. Specific operators and functions like $\ln$, $\cos$ and the d in differentials should be non-italic.

You may also use `$ 1/x $` for short in-line equations in the text like $1/x$. This is a reference to Equation 5.1.

Units

For dealing with dimensional values, the `siunitx` package is useful. In text or math mode a value can be assigned a unit with the following syntax:

```
\SI{value}{units}
```

An example is

```
\SI{2.0}{\micro\gram\per\cubic\metre}
```

resulting in the expression $2.0\,\mu\mathrm{g}\,\mathrm{m}^{-3}$. The package takes care of proper spacing between numbers and units as well as line breaks.

Bibliography

[1] Albert Einstein. Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig? *Annalen der Physik*, 323(13):639–641, 1905. DOI: 10.1002/andp.19053231314.